

Maximizing the Benefits of Zero-Energy Homes

by Ren Anderson, Rob Hammon, and Mike Keesee

There is an emerging trend in energy programs for new homes to shift from a focus on the use of individual high-performance components to a focus on overall home performance. Key examples of this trend include the MASCO Environments for Living program, the SMUD Zero-Energy Home program (see “Sacramento’s Zero-Energy Community,” *HE* Mar/Apr ’06, p. 24), the DOE Building America program, and California’s New Solar Homes Partnership (NSHP). The overall home performance benefits that these programs deliver include labor and cost benefits for builders and contractors; shared benefits from improvements in housing durability, reliability, and comfort; annual energy and utility bill savings for homeowners; and reductions in peak electric demand in new housing.

To help evaluate the effectiveness of this shift, we compared the first costs of two different approaches to increasing source energy savings beyond California’s Title 24 Residential Energy Code in a 2,592 ft² two-story home in Sacramento (see Figure 1). One program sought only to reduce space conditioning energy use by focusing only on improvements in HVAC equipment performance. The other program included all energy-saving options. The benefits of the whole-buildings approach are clear from comparing the costs and energy savings associated with the two programs. The program that targets only space conditioning energy use achieves maximum source energy savings of about 18% relative to Title 24 and has first costs that are



This Centex Homes model home in San Ramon, California, includes a range of second-generation energy efficiency features.

similar to the program that includes all available energy-saving options; only the all-options case saves more energy. Programs that target whole-building energy savings are more cost-effective for program providers and participants.

But how can a program planner choose which technologies should be included, or should be offered an incentive, in a given whole-house program? Candidate technologies must have the potential both to be acceptable to builders and to provide large incremental energy or cost savings in order to be considered for inclusion in above-code energy programs. Deciding which technologies will meet these criteria isn’t easy. For example, concentrating solar light pipes, radiant cooling, and even cool roofs may seem to be good candidates for inclusion in a given whole-house program—but in

fact, they may not always be the best choice (see “Ready for Prime Time?” p. 44).

Builders won’t implement a new technology unless it first meets minimum requirements for ease of integration with the residential construction process. Recognizing the relatively long time between initial market entry of a new technology and broad residential market adoption, it is important for energy program planners to carefully consider how far currently available technologies can increase performance before additional technologies are needed to fill remaining performance gaps leading to longer-term goals.

In view of the potential barriers to residential market transformation, a growing number of beyond-code residential energy programs in the

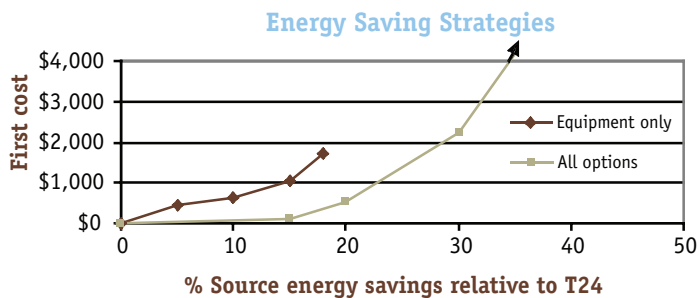


Figure 1. Comparison of energy-saving strategies for a two-story home in Sacramento.

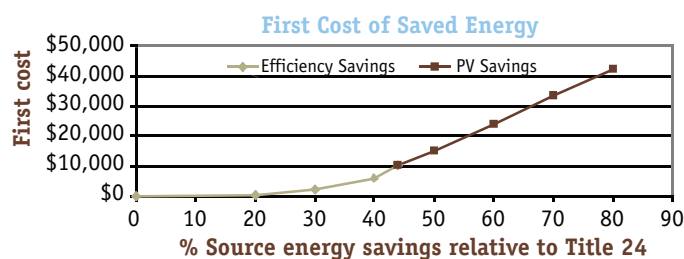


Figure 3. Average cost of total site energy savings.

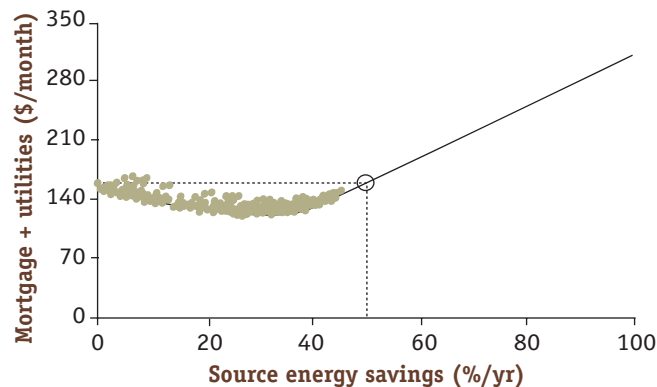


Figure 2. This least-cost curve for a two-story, 2,592-ft² home in Sacramento represents the maximum cost/performance potential that can be achieved based on the options used in the analysis. There are many points near the least-cost curve that perform nearly as well as points on the least-cost curve.

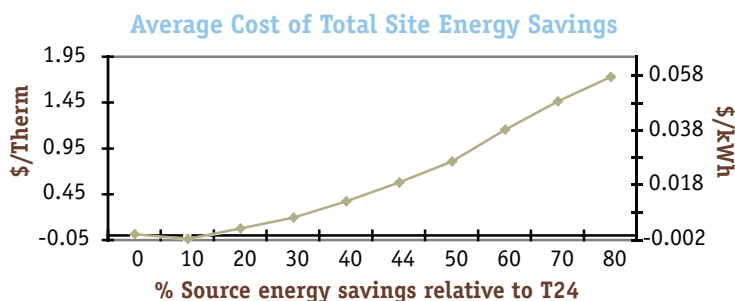


Figure 4. Incremental first costs associated with achieving different annual source energy savings levels.

United States are maximizing whole-house energy savings by

- providing technical support and incentives to builders to reduce the time required for new approaches to become standard practice; and by
- targeting multiple residential energy end uses to maximize overall benefits, rather than focusing energy-saving strategies on specific building components.

LEAST-COST APPROACH TO ZERO-ENERGY HOMES

Developing a consistent method for comparing the costs of energy savings realized by whole-building energy-saving options and interactions between different combinations of these options is a challenge. Researchers at the National Renewable Energy Laboratory (NREL) have developed the BEopt analysis method as one approach to solving this problem. BEopt determines the least-cost path to zero net energy (ZNE) homes, by evaluating the mar-

ginal costs of different combinations of energy efficiency and renewable-energy options. The results provide a starting point for developing multiyear program targets that can be updated based on actual program results. DOE's Building America program has recently adopted this method of analysis.

In California, programs designed to achieve a given level of energy savings cost effectively must also strive to minimize peak summer cooling demands. Residential cooling demands contribute significantly to high generation, transmission, and distribution (T&D) costs for California utilities. Sizable incentives have been added to Title 24 to encourage the development of construction approaches that minimize peak cooling loads in new homes. The cost of building new power plants and T&D systems can be considerably reduced if new homes are designed to have zero cooling demand during the times when peak cooling loads would normally occur. The BEopt analysis

method can also be used to identify the best ways to reduce cooling demand.

The BEopt analysis method uses an efficient sequential search optimization technique to find optimal and near-optimal combinations of discrete energy efficiency options. This approach involves searching all categories (wall type, ceiling type, window glass type, HVAC type, and so on) for the most cost-effective combination at each sequential point along the path to ZEH. Starting with the base case building, simulations are performed to evaluate (one at a time) all available options for improvement in the building envelope and equipment. Based on the results, the most cost-effective combination is selected as an optimal point on the path, and is put into a new building description. The process is repeated. At each step, the marginal cost of saved energy is calculated and compared with the cost of PV energy. From the point where further improvement in the building envelope or equipment has a higher marginal cost, the building

Ready for Prime Time?

Not all currently available technologies are suitable for use in residential programs that focus on realizing large incremental energy savings. Many successful products are energy neutral, provide modest energy benefits, or are targeted toward niche markets. There are four barriers to realizing large incremental energy savings in production homes:

- It may be difficult to scale the technology down to the footprint and capacities required for residential applications.
- Maintenance or operating requirements may be incompatible with use in residences, where technologies must operate for long periods of time with little or no maintenance or adjustment.
- The energy output or benefit of the technology may not match residential energy input needs.
- The technology may provide benefits other than large energy savings.

Here are some examples of technologies whose promise is apparent, but whose incremental energy savings contributions in new high performance homes may be limited.

Concentrating solar light pipes. A concentrating solar light pipe system is an example of a technology whose energy output does not match residential energy input needs. Homes have high perimeter-total floor area ratios and window areas that provide lots of natural lighting. In addition, most residential lighting use occurs during the night, when solar lighting is not available. Areas like closets that do not have access to natural lighting do not contribute much to total residential lighting use. This combination of factors makes it difficult for a solar light pipe system to provide high value to builders or consumers from an energy savings perspective. The technology must meet minimum builder and contractor performance requirements if it is to be used in new homes, and in order for the technology to be included in planning for future residential energy programs.

Active solar-thermal space-heating systems. Active solar-thermal space-heating systems face a similar mismatch between energy output and residential input needs. The solar-thermal resource is available primarily in the summer, when residential space heating is not required. Seasonal storage systems have been proposed as a possible future solution to this problem.

Cool-roof materials. Cool-roof materials have the potential to reduce roof temperatures, reduce the rate of thermal degradation, and increase the lifetime of some roofing materials. However, because there is little or no thermal connection between the roof and the living space of a high-performance new home with a ventilated tile roof, high levels of ceiling insulation, and ducts in the conditioned space, cool-roof materials are not currently expected to realize significant energy savings in high-performance new homes. Over the long term, as ceiling insulation levels continue to increase and cool-roof technologies continue to mature, incremental energy savings from cool-roof materials may begin to compete with the savings achieved by increased levels of ceiling insulation.

Radiant cooling systems. One of the major dollar savings realized with increases in house performance is the reduced cost of the furnace and air conditioning systems. This is due to corresponding reductions in peak space conditioning load and space conditioning system size. As envelope performance increases in a 2,592-ft² house in Sacramento, for example, the size of the air conditioner that is required to meet peak cooling loads is reduced from 3 tons to 1.5 tons. If an advanced cooling technology cannot be cost-effectively provided by a system in the 1.5-ton range, then that technology will not meet minimum builder and contractor performance requirements for use in new homes. A radiant floor- or radiant ceiling-based cooling system is an example of a cooling technology with significant first costs for large-area radiant coils that cannot be reduced in size as cooling capacity is reduced.

design is held constant, and PV capacity is increased to reach ZNE.

By using BEopt, we found that it is possible to build a two-story, 2,592-ft² home in Sacramento, at an incremental cost of about 5% relative to a standard Title 24 home, that will reduce total annual heating energy use by 70%, reduce annual cooling energy use by 60%, and reduce total source energy use by 60% (see “BEopt Assumptions,” p. 46). Increases in the energy efficiency of the reference home provide 44% of the annual energy savings, and 1.5 kW of the peak demand savings from cooling. A west-facing 2 kilowatt peak DC (kW_{PDC}) PV system with thermostat setback is added to the base efficiency package to achieve nearly zero peak demand from cooling.

In choosing the package of least-cost energy options for this house, we were careful to select ones that production builders would find acceptable. However, many of these technology options still face significant barriers to broad use before they can move to higher levels of market maturity. For example, the use of high levels of wall and ceiling insulation also requires significant changes in design and construction practices, including the use of advanced framing techniques such as 2 x 6 framing and raised-heel roof trusses. And action on the part of a home's occupants is necessary in order for a setback thermostat to provide reliable demand reductions during peak cooling days.

BASE CASE BUILDING CHARACTERISTICS

Incremental energy savings and incremental home costs are calculated relative to a base case building that meets Title 24 requirements for Sacramento (California climate zone 12). The base case building includes low-leakage tested ducts with R-4.2 insulation, low solar heat gain windows, R-3 wall insulation, R-38 ceiling insulation, and a 13-SEER (seasonal energy efficiency ratio) air

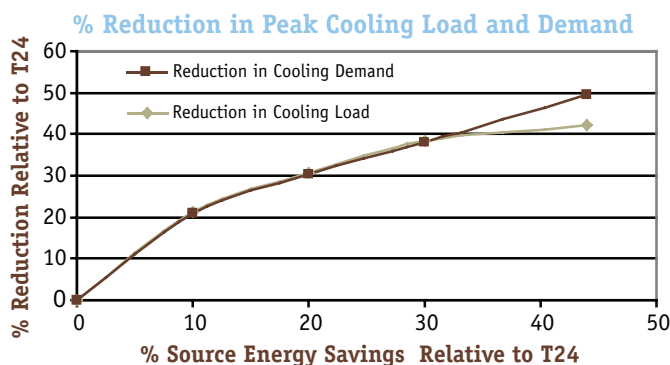


Figure 5. Impact of energy efficiency improvements on peak cooling loads and peak cooling demand.

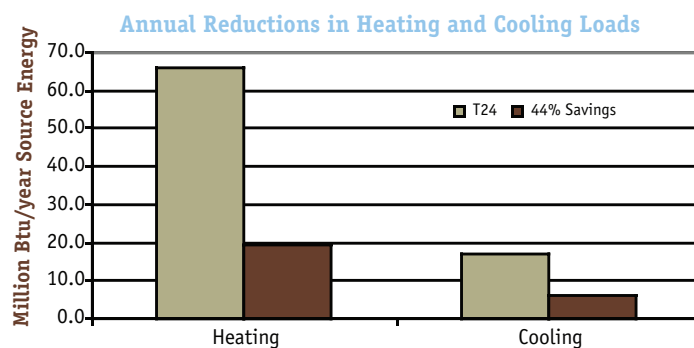


Figure 7. Annual heating and cooling source energy use can be significantly reduced even before the addition of a PV system.

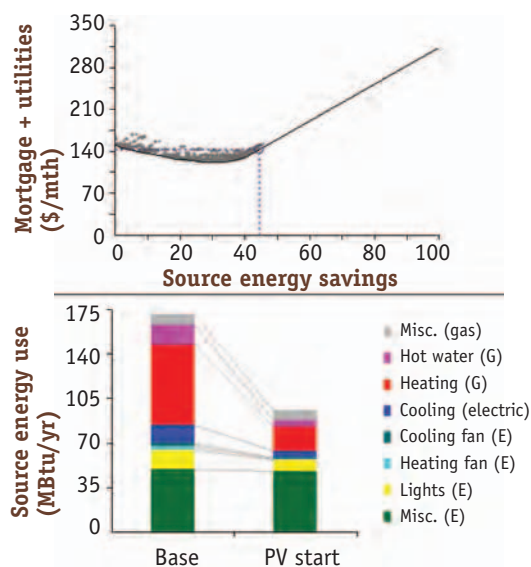


Figure 6. Annual energy end uses at the 44% source energy savings level.

conditioning system. The occupancy and operational assumptions used in the study are defined in the Building America Research Benchmark and include time-of-day profiles for occupancy, appliance and plug loads, lighting, domestic hot water use, ventilation, and thermostat settings.

To provide the reader with a closer look at the analysis results, the least-cost curve is shown as a function of source energy savings in Figure 2 on p. 43. At zero source energy savings, the cost on the curve at the vertical axis represents the annual utility bill for a homeowner with a Title 24 home. Each point on the curve represents a different combination of equipment and envelope options for the home. The BEopt method

conducts an annual energy simulation for a large number of possible combinations of options in the vicinity of the least-cost curve. The least-cost curve represents the lower bound of these combinations of options.

One of the benefits of the BEopt analysis method is that it can identify combinations of options represented by the points nearest the least-cost curve that have nearly equivalent cost and performance. The marginal cost of increased energy efficiency is equal to the current marginal cost of electricity from residential PV when source energy

savings reach about 44%. The straight line that begins at source energy savings of 44% represents the cost of using a net metered, grid-connected PV system to meet remaining home energy needs. The neutral cost point where total energy-related homeowner costs for a high-performance home are equal to the initial utility costs for a Title 24 home occurs at source energy savings of about 50%.

The first costs associated with achieving different levels of source energy savings on the least-cost curve are shown in Figure 3 on p. 43. A home with 40% in annual source energy savings relative to Title 24 increases first costs by \$5,000; a 50% home increases costs by \$15,000; and a 60% home increases costs by \$25,000.

If the base home price for the Title 24 home is \$500,000, the corresponding fractional increases in home cost required to achieve these higher performance levels are 1%, 3%, and 5% respectively.

The average 30-year cost of site energy savings is shown as a function of source energy savings level in Figure 4. The first cost required to achieve each savings level was divided by the total site energy savings for gas and electric end uses for the 30-year period of the current study to determine the average costs shown in Figure 4 on p. 43. The average cost of saved energy compares favorably with current residential energy costs out to a source energy savings level of about 50%–60%. The average cost of electricity provided by PV is about \$0.17/kWh.

Changes in peak cooling load and peak cooling demand for the base case home are shown as a function of source energy savings in Figure 5. Improvements in home energy efficiency reduce peak cooling load by over 40% (lower curve in Figure 5) and peak cooling demand by 50% (upper curve in Figure 5). The additional reduction in demand relative to load is the result of the increase of air conditioner SEER from 13 to 15 at the 44% source energy savings level. In the current study, which assumes installed PV costs of \$7.50/W, the marginal cost of electric savings from PV is less than the marginal cost of electric savings from high-SEER air conditioning systems for source energy savings levels greater than 44%.

The development of the BEopt analysis method was influenced by several factors. First, the method identifies intermediate optimal points all along the path of interest (that is, minimum-cost building designs at different target energy savings levels), not just the global optimum or the zero net energy (ZNE) optimum. Second, the method allows discrete rather than continuous building options to be evaluated, reflecting realistic construction options. Third, an additional benefit of the search strategy is the identification of near-optimal alternative designs along the path, allowing for substitution of essentially equivalent solutions based on builder or contractor preferences.

Base Case Building Characteristics

A simple two-story 2,592-ft² home in Sacramento with an attached two-car garage was used as the model for this study. The home has 1-foot eaves and a slab foundation. Window area was assumed to be 18% of floor area and was equally distributed among outside walls. Window distribution is a user input and can be modified to reflect specific home designs. The study was limited to one orientation (back facing west), allowing full exposure of the windows on the back of the home to afternoon sun. Adjacent homes 10 feet to the north and south provide shading of sidewalls. The energy options considered in the study include space conditioning systems, envelope systems, hot water systems (including tankless and solar hot water), lighting systems, major appliances, and grid-connected residential PV. No specific options that address miscellaneous electric loads other than major appliances were included in the determination of the base case building design. The homeowner costs calculated in the study assume a 30-year mortgage at a 7% interest rate with a 3% general inflation rate and a 5% real discount rate. The net present value of replacements for options with lifetimes of less than 30 years were included in option costs.

Cost Assumptions

Each option has an assumed first cost and lifetime. Costs used in the analysis represent retail costs and include national average estimated costs for hardware, installation labor, overhead, and profit. Construction costs (wall insulation, ceiling insulation, foundation insulation, and so on) are typically based on national average cost data. Window and HVAC costs are based on quotes from manufacturers' distributors. Appliance costs are based on manufacturers' suggested retail prices.

Building construction options (wall insulation, ceiling insulation, foundation insulation, windows, and so on) are assumed to have a 30-year lifetime. Equipment and appliance options typically have a 10- or 15-year lifetime. Lifetimes for lighting options (incandescents and CFLs) are modeled based on cumulative hours of use.

Utility costs are assumed to escalate at the rate of inflation (that is, to be constant in real terms). The on-site power option used for this study was a residential PV system with an installed cost of \$7.50 per peak watt of DC power including present value of future operation and maintenance (O&M) costs. Natural gas is assumed to have an average cost of \$1 per therm. Electricity is assumed to have an average cost of \$0.127/kWh. These costs were chosen as representative based on energy costs over the past several years. The impact of recent or potential longer-term increases in the cost of energy was not considered in the current study. The home is assumed to have a gas water heater or a solar hot water heater with gas backup, a gas furnace, a gas clothes dryer, and a gas stove. (A gas clothes dryer was chosen based on the goal of defining a building that minimizes peak electric demand.) All cost assumptions are user inputs that can be modified to reflect actual costs in specific projects.

The cost estimates used in this study do not include the initial costs required to reengineer home designs, state and local financial incentives and rebates, or other builder costs, such as warranty and callback costs. All of these additional costs may have a significant impact on builder business decisions related to the construction of new home designs. These costs must be considered as part of the design of programs aimed at increasing the construction of high-performance homes.

The nominal energy efficiency ratio (EER) of the 13-SEER air conditioning system is assumed to be 11. At the 60% savings level, a 2kW_{DC} PV array with a 25% derate factor provides the 1.5 kilowatts required to meet peak cooling demand.

The annual energy end uses in the home at the 44% savings point relative to Title 24 where the incremental cost of additional energy efficiency savings equals the cost of energy savings from PV is shown in Figure 6. The right-hand side of Figure 6 also includes a summary of the options used to increase the efficiency of the home relative to the base Title 24 home.

In Figure 7, the cooling electric demand for the Title 24 reference house is compared to that for the house with 44% annual source energy savings relative to Title 24 for a peak cooling day with a thermostat setpoint of 76°F. Even without added PV, the energy efficiency options that have been implemented in the 44% house reduce peak cooling demand by about 1.5 kilowatts. In addition to the reductions in peak cooling demand, the energy efficiency upgrades in the 44% house also reduce total annual heating energy use by 70% and annual cooling energy use by 60%.

Because of the large reduction in the cooling load of the 44% house, a simple thermostat setback and setup strategy can have a significant impact on cooling demand in the high-performance home. To achieve the cooling demand results shown in Figure 8, the cooling setpoint was set at 76°F from 10 pm to 10 am. The setpoint was reduced from 76°F to 72°F from 10 am to 4 pm and then increased from 72°F to 78°F from 4 pm to 10 pm. Precooling strategies have previously been shown to be cost-effective alternatives for commercial buildings. This simple strategy shifts the cooling peak in the 44% house back from late afternoon to midafternoon (lower curve in Figure 8). The same strategy also provides a slight shift in the Title 24 house, but is insufficient to reduce the late-afternoon peak cooling demand (upper curve in Figure 8).



To succeed, a near-zero-energy home must excel in all aspects of home performance. Notice the outside air intakes neatly tucked under the second-story eave in this Lone Star Ranch home in Frisco, Texas.

The impact of adding a west-facing 2 kW_{DC} PV system in combination with a thermostat setback and setup strategy is shown in Figure 9. The net impact of the PV system (in combination with energy efficiency measures and the thermostat setback/setup strategy) is to reduce peak cooling demand to nearly zero throughout most of the day in the 44% house (lower curve in Figure 9). The 2 kW PV system reduces the peak cooling demand in the Title 24 house in the late morning and early afternoon, but has little impact on the late-afternoon peak (upper curve in Figure 9).

By maximizing the investment in energy efficiency options before making investments in advanced options like PV, the size and therefore the cost, of the investment in the advanced option is minimized. A 4 kW_{DC} array at a cost of \$30,000 would be required to reduce peak cooling demand on the grid to zero if no additional investment in energy efficiency beyond Title 24 were made. Not only would a PV-only approach cost more than the optimum found using the BEopt analysis method (\$30,000 versus \$25,000), but it would also provide significantly lower aggregate peak and off-peak energy savings than the integrated energy efficiency and renewable-energy approach implemented in BEopt's least-cost optimization engine.

BEopt provides a way for residential program planners to identify the residential technologies that can be successfully used by production builders and that also have the potential to provide large incremental energy savings relative to currently available technologies. By focusing attention on the options that are most likely to contribute to future energy savings, this screening approach also identifies features of advanced technologies that require additional development to increase the likelihood of their future success.



Ren Anderson is leader of the Residential Research Section in the National Renewable Energy Laboratory (NREL) Buildings and Thermal Systems Center; Rob Hammon is principal of ConSol, an energy services provider for production home builders, located in Stockton, California; and Mike Keese is the PV project manager for the Sacramento Municipal Utility District.

The authors gratefully acknowledge the support of DOE, the Sacramento Municipal Utility District, and the California Energy Commission for portions of this work.

Impact of Setback/Setup Thermostat Strategy on Cooling Demand During August 20 Peak Day

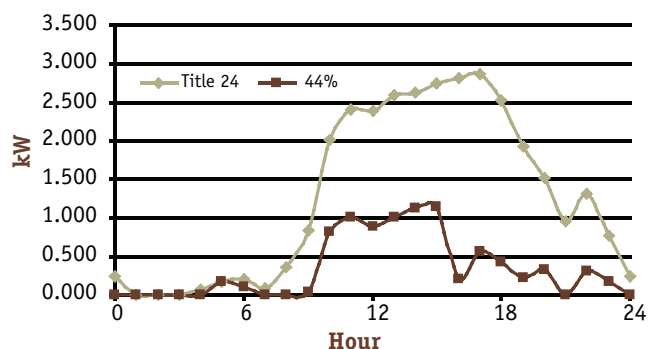


Figure 8. Thermostat setback and setup significantly reduce peak cooling demand in the 44% house.

Peak Cooling Demand with Setup/Setback and a 2 kW PV System during August 20 Peak Day

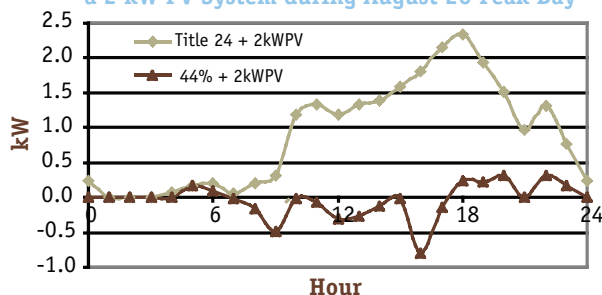


Figure 9. A 2 kW west-facing PV system has little impact on peak cooling demand in the T24 house (upper line) and reduces cooling demand to nearly zero in the 44% house (lower line).

For more information:

Ren Anderson
NREL
1617 Cole Blvd.
Golden, CO 80401
E-mail: ren_anderson@nrel.gov

Rob Hammon
ConSol
7407 Tam O'Shanter Dr.
Stockton, CA 95210
E-mail: rob@consol.ws

Mike Keese
SMUD
6201 S St.
MS# B257
Sacramento, CA 95817
E-mail: mkeese@smud.org